



OPEN SOURCE • RUST

Context for AI coding agents, in one binary.

Code and specs indexed into a single graph.
Searched, traced and served **within a token budget.**

swipe →



THE PROBLEM

Agents read **files**. Projects are made of **relationships**.

- Whole files get dumped into the prompt — tokens burn on code the task never needed.
- Specs and code live apart — which function satisfies REQ-105?
- Impact is guesswork — what breaks if I change this symbol?
- External indexers mean daemons, interpreters and setup.

THE IDEA

One graph for **specs + code.**

Nodes

Requirement · Task · ADR ·
DocSection · Symbol · File

Edges

Satisfies · DependsOn ·
Implements · DefinedIn

```
// @spec REQ-105
fn compact_delta(base, delta) { ... }

$ nexspec trace REQ-105
depth=0 Requirement  Two-layer CSR
depth=1 Satisfies    compact_delta
depth=2 DefinedIn    src/graph/csr.rs
```

Git in, **consistent** indexes out.

Git diff

gix • native tree diff, no subprocess



Tree-sitter

TS/JS • Python • Go • Rust

Comrak

.specs/*.md → REQ / TASK / ADR



Sync Coordinator

WAL • staging • atomic sync_version



redb

metadata

CSR

topology

Tantivy

BM25

HNSW

vectors

From a query to a **token budget**.

1**Hybrid search**

BM25 (Tantivy) + vectors (HNSW, optional)

2**RRF fusion**

Reciprocal Rank Fusion picks the seeds

3**k-hop expansion**

Walk the graph for related specs and symbols

4**AST pruning**

Bodies become { ... }, signatures stay

5**Budget + serialize**

Deterministic cut, 90% margin, dense Markdown

DESIGN CHOICES

Built to **never** serve a half-written index.

- **Crash-safe.** Every write goes through one coordinator: WAL first, atomic rename after. --resume recovers deterministically.
- **Lock-free reads.** Immutable mmap base + copy-on-write delta. Readers never block on a sync.
- **Lean when you need it.** The vector engine is optional and lazy. CI builds skip it entirely.
- **Incremental.** Only what changed in Git gets reprocessed.

USING IT

A CLI for you. An **MCP** server for your agent.

```
nexspec init  
nexspec sync  
nexspec search "auth" --max-tokens 4000  
nexspec trace REQ-105  
nexspec blame compact_delta  
nexspec diff --staged  
nexspec mcp      # stdio, for Claude Code, Cursor...
```

query_context

trace_requirement

find_impacted_code

get_symbol_history

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Try it. **Break it.** Tell me what's missing.

Written in Rust. Built spec-first, with the specs as the source of truth.

github.com/leandroluk/rust-nexspec

leandroluk.github.io/rust-nexspec